Chapter 4

Diode Characteristics & Mathematical Models

4.1 Introduction

In the previous lecture, we established the physics of the PN junction: how doping creates P and N regions, how diffusion creates a depletion region, and how a potential barrier is formed. Now, we transition from physics to engineering application.

A **diode** is the simplest two-terminal electronic component. Its defining characteristic is that it acts as a ”one-way valve” for electricity. However, unlike a perfect switch, a real diode has non-linear characteristics—it has a ”knee” voltage, internal resistance, and breakdown limits. Understanding these **non-ideal characteristics** is crucial for designing reliable circuits.

Learning Objectives

By the end of this chapter, you should be able to:

- Sketch and analyze the V-I Characteristic Curve of a diode.
- Define Knee Voltage (V_K) and Breakdown Voltage (V_{BR}).
- Apply the Shockley Diode Equation to calculate current.
- Distinguish between DC (Static) and AC (Dynamic) resistance.
- Use Equivalent Circuit Models (Ideal, Practical, Complete) for circuit analysis.

4.2 The V-I Characteristic Curve

The voltage-current (V-I) characteristic curve is the graphical "fingerprint" of a diode. It plots the voltage across the diode (V_D) on the x-axis versus the current flowing through it (I_D) on the y-axis.

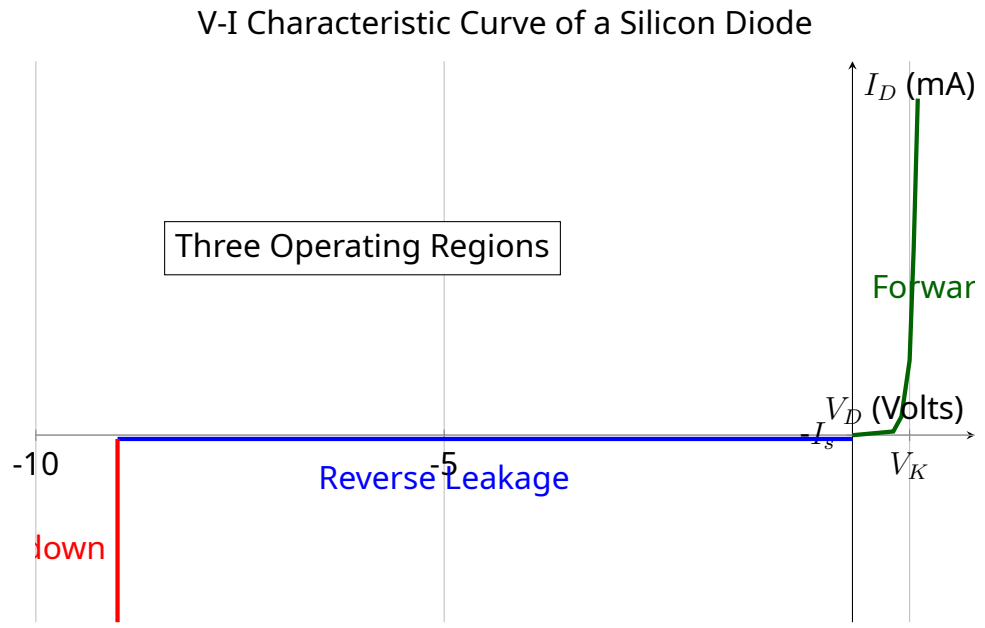


Figure 4.1: The V-I Characteristic Curve showing Forward, Reverse, and Breakdown regions.

The curve reveals three distinct regions of operation:

4.2.1 Forward Bias Region ($V_D > 0$)

- **0V to Cut-in Voltage (V_K):** When the applied voltage is small (< 0.5V), it is insufficient to overcome the barrier potential. The depletion region narrows but does not vanish. Current is negligible.
- **Cut-in / Knee Voltage (V_K):** As voltage approaches the barrier potential, current begins to rise. Once V_D exceeds this threshold, the depletion region collapses, and majority carriers rush across the junction.
  - **Silicon (Si):** $V_K \approx 0.7$ V.
  - **Germanium (Ge):** $V_K \approx 0.3$ V.
- Above V_K: The current rises exponentially. A tiny increase in voltage causes a massive spike in current. The diode behaves like a closed switch with a small voltage drop.

4.2.2 Reverse Bias Region ($V_D < 0$)

- Ideally, the diode acts as an open switch ($I_D = 0$).
- In reality, a tiny **Reverse Saturation Current (I_S)** flows due to minority carriers (thermally generated).
- For Silicon, I_S is typically in the nano-ampere range (10^{-9} A).

4.2.3 Breakdown Region

If the reverse voltage exceeds the **Peak Inverse Voltage (PIV)** or Breakdown Voltage (V_{BR}), the diode fails.

- **Avalanche Breakdown:** Minority carriers gain enough kinetic energy to knock valence electrons out of atoms, creating a chain reaction.
- Current shoots up rapidly. Unless limited by a resistor, the diode will be destroyed.

4.3 Mathematical Model: The Shockley Equation

The behavior of the PN junction is modeled by the **Shockley Diode Equation**:

$$I_D = I_S \left(e^{\frac{V_D}{nV_T}} - 1 \right) \tag{4.1}$$

Where:

- **I_D**: Diode Current (A).
- **I_S**: Reverse Saturation Current (A).
- **V_D**: Applied Voltage across the diode (V).
- **n**: Ideality Factor (1 for ideal/Ge, 2 for Si at low currents). We usually assume $n = 1$.
- **V_T**: Thermal Voltage (V).

4.3.1 Thermal Voltage (V_T)

The Thermal Voltage represents the energy of electrons due to temperature:

$$V_T = \frac{kT}{q} \tag{4.2}$$

At room temperature ($25°$C or 298 K):

$$V_T \approx 0.026 \text{ V} = 26 \text{ mV}$$

Example 1: Calculating Diode Current

A Silicon diode has a saturation current $I_S = 10$ nA (10×10^{-9} A). Calculate the current at a forward voltage of 0.6 V at room temperature ($V_T = 26$ mV, $n = 1$).

Solution: Using the approximate forward bias equation ($e^x \gg 1$):

$$I_D \approx I_S e^{\frac{V_D}{nV_T}}$$

$$I_D = 10 \times 10^{-9} \times e^{\frac{0.6}{0.026}}$$

$$I_D = 10^{-8} \times e^{23.077}$$

$$I_D \approx 10^{-8} \times (1.05 \times 10^{10}) = 105 \text{ mA}$$

4.4 Diode Resistances

Since the diode is non-linear, it does not have a single fixed resistance value.

4.4.1 DC (Static) Resistance (R_D)

This is the resistance offered to a constant DC current.

$$R_D = \frac{V_D}{I_D} \tag{4.3}$$

- Near the knee (0.7V), resistance is low ($10 - 50\Omega$).
- In reverse bias, resistance is very high ($M\Omega$).

4.4.2 AC (Dynamic) Resistance (r_d)

This is the resistance offered to a changing (AC) signal. It is the reciprocal of the slope of the tangent line at the Q-point.

$$r_d = \frac{dV_D}{dI_D}$$

By differentiating the Shockley equation, we get a simple approximation at room temperature:

$$r_d = \frac{26 \text{ mV}}{I_D} \tag{4.4}$$

4.5 Diode Equivalent Circuit Models

To analyze diodes in circuits, we replace the physical symbol with a model composed of standard linear components.

4.5.1 The Ideal Diode Model

Used for quick mental checks.

- **Forward:** Closed Switch ($V_D = 0$).
- **Reverse:** Open Switch ($I_D = 0$).

4.5.2 The Practical Model

Standard for general design. Accounts for the barrier potential.

- **Forward:** Closed Switch + **0.7V Battery**.
- **Reverse:** Open Switch.

4.5.3 The Complete Model

Used for precision simulation.

- Forward: Switch + 0.7V Battery + Small Resistor (r_d).
- Reverse: Large Resistor (R_{rev}) || Current Source (I_S).

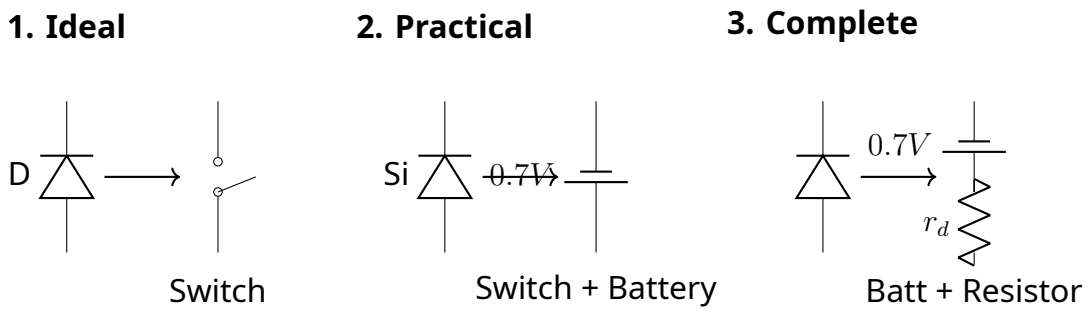


Figure 4.2: Equivalent Circuit Models for a Diode in Forward Bias.

4.6 Review Questions

Part A: Conceptual Questions

1. Sketch the V-I characteristic curve of a silicon diode. Label the knee voltage, breakdown voltage, and leakage current region.
2. Explain the physical significance of the "Ideality Factor" (n) in the Shockley equation.
3. Differentiate between Static Resistance (R_D) and Dynamic Resistance (r_d). Which one is used for small-signal analysis?
4. Why does the forward voltage drop of a diode decrease as temperature increases?
5. Describe the difference between the Ideal Diode Model and the Practical Diode Model. When would you use each?
6. What is "Avalanche Breakdown"? How does it differ from Zener Breakdown?
7. Define "Thermal Voltage" (V_T). What is its approximate value at room temperature?
8. Why is the reverse saturation current (I_S) temperature-dependent?
9. What is the effect of Diffusion Capacitance (C_D) on the switching speed of a diode?

10. Explain why a diode in reverse bias acts like a capacitor (Transition Capacitance C_T).

Part B: Advanced Calculations & Analysis

11. Current Calculation: A Germanium diode ($n = 1, I_S = 1\mu$A) conducts 5 mA of current. Calculate the forward voltage drop across it at $27°$C.

12. Dynamic Resistance: Calculate the AC resistance (r_d) of a diode operating at a forward current of (a) 1 mA and (b) 100 mA. Comment on the result.

13. Model Analysis: A circuit consists of a 10V DC source, a silicon diode, and a $1k\Omega$ resistor in series. Calculate the circuit current using:

    (a) The Ideal Model.

    (b) The Practical Model ($V_K = 0.7$V).

    (c) Calculate the

14. Temperature Effect: A silicon diode has a forward drop of 0.7V at $25°$C. If the temperature coefficient is -2.5 mV/°C, estimate the forward voltage drop at $100°$C (boiling water).

15. Reverse Leakage: The reverse saturation current of a diode doubles for every $10°$C rise in temperature. If $I_S = 10$ nA at $25°$C, calculate the leakage current at $85°$C.